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Interactive conjoint environment

Abstract

An interactive vehicle conjoint study survey system includes a first virtual machine including an application server operating a management program generating, modifying and saving conjoint environment survey content for access by a survey taker. A choice task content list is generated by the management program and is reviewed and completed by the survey taker. A reader program of the first virtual machine identifies survey taker choices selected from the choice task content list to be displayed. A survey taker interface is formatted to conform to an interface window completely displayed on a screen of a survey taker device defining a smart phone displays the survey taker choices.

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Background/Summary

- (1) The present disclosure relates to systems and methods to generate and refine vehicle data conjoint surveys.
- (2) Current vehicle conjoint study surveys are used by vehicle designers during a vehicle design phase to understand the voice of prospective customers and how they make tradeoffs between item choices including product features and feature prices for vehicle manufactures to price features and to help determine features to be included in different vehicles. Current vehicle conjoint study surveys are conducted on-line using personal computers (PC)s. Presentations are commonly administered by third party contractors or vendors who identify potential survey takers and use computer station monitors and lap-top computer screens which are large enough to present multiple volumes of vehicle data. Current vehicle conjoint study surveys, however, do not generalize well when deployed on small-screen mobile devices. In-survey quality may also be diminished by survey taker behavior known to diminish response quality.
- (3) Survey taker behavior impacts survey completion rates which may be diminished due to survey taker early drop-out and includes reduced choice tasks dwell time and tap and drag actions. Current vehicle conjoint studies lack placement and frequency of sensory feedback. Current vehicle conjoint studies also suffer from survey taker task comprehension resulting in survey taker boredom, diminished attention, and reduced choice task engagement. At present, survey takers cannot participate in conjoint study surveys using survey taker preferred devices including mobile devices such as mobile phones because current vehicle conjoint studies now presented on large displays such as on personal computer screens and monitors do not generalize well on known mobile devices, therefore limiting the possible pool size of prospective survey takers.
- (4) Thus, while current systems and methods to perform vehicle conjoint studies achieve their intended purpose, there is a need for a new and improved system and method to generate and actively refine vehicle conjoint studies.

SUMMARY

- (5) According to several aspects, an interactive conjoint study survey system includes a first virtual machine including an application server operating a management program generating, modifying and saving a conjoint environment survey content for access by a survey taker. A choice task content list is generated by the management program and is reviewed and completed by the survey taker. A reader program of the first virtual machine identifies multiple survey taker choices selected from the choice task content list to be displayed. A survey taker interface displays the multiple survey taker choices, the survey taker interface being formatted to conform to an interface window completely displayed on a screen of a survey taker device defining a smart phone.
- (6) In another aspect of the present disclosure, a vendor program database collects information for a conjoint environment survey of data including data provided by a manufacturer enabling the survey taker via a programming function to access the first virtual machine.
- (7) In another aspect of the present disclosure, a second virtual machine forwards data requests to and retrieves conjoint environment survey data from the second virtual machine.
- (8) In another aspect of the present disclosure, the second virtual machine includes a database of developmental vehicle data for a vehicle under development including a vehicle attribute list and descriptions of features and options available in the vehicle under development and listed on the choice task content list.
- (9) In another aspect of the present disclosure, a data collection link includes camera image data and facial data, wherein applying input data of the survey taker via the data collection link including multiple survey taker camera image data and multiple survey taker facial data, the API using the management program identifies when and if a communication signal as a wireless or a wired transfer of data is forwarded to the survey taker device.
- (10) In another aspect of the present disclosure, a device notification is generated in response to the communication signal and is presented on the interface window or on the survey taker device, the

device notification including at least one of a haptic notification and an indicator light including a blinking or a steady light on the interface window or on the survey taker device to notify the survey taker that additional attention needs to be paid to the conjoint environment survey content.

(11) In another aspect of the present disclosure, if the management program identifies the survey taker has met a predetermined threshold of distraction, boredom, eye-vision wander, response error, or inattention, the device notification defines a rest screen.

(12) In another aspect of the present disclosure, an application programming interface (API) of the first virtual machine receives information including but not limited to the multiple survey taker choices, a survey taker focus status, survey taker eye-tracking data, timing or latency data for the survey taker to respond to data choices presented, indication of survey taker distraction or an indication of survey taker misunderstanding of a conjoint environment survey question such as repetition of a same response.

(13) In another aspect of the present disclosure, a first camera is provided for the survey taker device, the management program receiving data from the first camera, wherein camera images and survey taker features are applied using the management program to monitor eye movement of the survey taker to define eye-tracking data indicating survey taker attentiveness or distraction from accomplishing the survey, and visual cues of the survey taker including facial features representing survey taker distraction.

(14) In another aspect of the present disclosure, the survey taker interface is further formatted to conform to being displayed on a viewing screen of a personal computer (PC).

(15) According to several aspects, a method to perform a vehicle interactive conjoint study survey comprises: operating a management program to generate, modify and save conjoint environment survey content for access by a survey taker; generating a choice task content list using the management program to be reviewed and completed by the survey taker; sending information including multiple survey taker choices, a survey taker focus status, survey taker eye-tracking data, timing or latency data for the survey taker to respond to data choices presented, indication of survey taker distraction or an indication of survey taker misunderstanding of a conjoint environment survey question such as repetition of a same response to an application programming interface (API); monitoring the survey taker with the API and using the management program to identify when and if a communication signal is forwarded to a survey taker device; and generating a device notification in response to the communication signal and presenting the device notification on the survey taker device.

(16) In another aspect of the present disclosure, the method further includes displaying the multiple survey taker choices on a survey taker interface of the survey taker device.

(17) In another aspect of the present disclosure, the method further includes formatting the survey taker interface to conform to an interface window completely displayed on a screen of the survey taker device defining a smart phone.

(18) In another aspect of the present disclosure, the method further includes providing the survey taker interface with multiple choice tasks for selection by the survey taker to retain an interest and concentration of the survey taker including at least one of: identifying the multiple choice tasks on a wedge-choice wheel presented in a wedge design having a circular wheel design; identifying the multiple choice tasks on an angular-choice wheel having angular rings in a circular wheel design with features presented on multiple wedges; and identifying the multiple choice tasks on a polygon-shaped choice wheel having different polygons presented on vertices having annular rings with individual ones of the annular rings successively numbered.

(19) In another aspect of the present disclosure, the method further includes applying camera image data and facial data as input data of the survey taker.

(20) In another aspect of the present disclosure, the method further includes: generating the device notification as a haptic notification or an indicator light including a blinking or a steady light on the survey taker device to notify the survey taker that additional attention needs to be paid to the

conjoint environment survey program; and conforming the device notification as a rest screen if the management program identifies the survey taker has met a predetermined threshold of distraction, boredom, eye-vision wander, response error or inattention.

(21) In another aspect of the present disclosure, the method further includes actuating a reader program to identify survey taker choices selected from the choice task content list to be displayed.

(22) According to several aspects, a method to perform a vehicle interactive conjoint study survey comprises: operating a management program to generate, modify and save a conjoint environment survey content for access by a survey taker; generating a choice task content list using the management program to be reviewed and completed by the survey taker; monitoring eye movement of the survey taker to define eye-tracking data indicating survey taker attentiveness or distraction from accomplishing a survey, and evaluating visual cues of the survey taker including facial features representing survey taker distraction and using the management program to identify when and if a communication signal is forwarded to a survey taker device; creating in-survey task breaks in response to measured behaviors of the survey taker to allow time off for the survey taker to relax prior to completing the survey; measuring survey taker in-survey focused attention and choice task survey taker engagement; and generating a device notification and presenting the device notification on the survey taker device.

(23) In another aspect of the present disclosure, the method further includes sending survey user information including survey taker selections, a survey taker focus status, survey taker eye-tracking data, timing or latency data for the survey taker to respond to data choices presented, indication of survey taker distraction or an indication of survey taker misunderstanding of a conjoint environment survey question such as repetition of a same response to an application programming interface (API).

(24) In another aspect of the present disclosure, the method further includes measuring a survey user response pattern including the survey taker repeatedly selecting a first-choice profile regardless of option content, eye-tracking including prolonged eye movement away from a device interface window, and interactive behaviors indicative of reduced attention including reduced dwell time in responding to choices.

(25) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.
- (2) FIG. 1 is a flow diagram of a vehicle interactive conjoint study survey system according to an exemplary aspect;
- (3) FIG. 2 is a front elevational view of a task screen defining a wedge-choice wheel identifying multiple choice tasks for the system of FIG. 1;
- (4) FIG. 3 is a front elevational view of a task screen defining an angular-choice wheel identifying multiple choice tasks presented having angular rings for the system of FIG. 1; and
- (5) FIG. 4 is a front elevational view of a task screen defining a wedge-choice wheel identifying multiple choice tasks for the system of FIG. 1.

DETAILED DESCRIPTION

- (6) The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.
- (7) Referring to FIG. 1, an interactive conjoint study survey system **10** provides a vendor program

database having a home page **12** collecting information from which a vendor conducting a conjoint environment survey of vehicle data such as data provided by an automobile manufacturer enables a survey taker **14** via a programming function **16** to access a first virtual machine **18**. The first virtual machine **18** includes an application server **20** operating a management program **22** which generates, modifies and saves a conjoint environment survey content for access by the survey taker **14** and the vendor. The management program **22** generates a choice task content list **24** to be reviewed and completed by the survey taker **14**. The first virtual machine **18** forwards data requests to and retrieves vehicle conjoint environment survey data from a second virtual machine **26**. The second virtual machine **26** includes a database **28** of developmental vehicle data for a vehicle **30** under development including a vehicle attribute list and descriptions of features and options which may be available in the vehicle being designed and which are listed on the choice task content list **24**.

(8) A reader program **32** of the first virtual machine **18** identifies multiple survey taker choices selected from the choice task content list **24** to be displayed which are forwarded by a choice code **34** to a survey taker interface **36**. The survey taker interface **36** is formatted to conform to an interface window **38** which may be completely displayed on a screen of a survey taker device **40** shown such as a smart phone or displayed on a viewing screen of a personal computer (PC) **42** such as a PC monitor or a lap-top computer viewing screen also shown.

(9) The computer including the PC **42** described in reference to FIG. **1** is a non-generalized, electronic control device having a preprogrammed digital controller or processor, memory or non-transitory computer readable medium used to store data such as control logic, software applications, instructions, computer code, data, lookup tables, etc., and a transceiver or input/output ports. The computer readable medium includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. The non-transitory computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. The non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device. Computer code includes any type of program code, including source code, object code, and executable code.

(10) An application programming interface (API) **44** of the first virtual machine **18** via a data collection link **46** which may be transferred wirelessly or via a wire connection receives information including but not limited to survey taker selections, a survey taker focus status, survey taker eye-tracking data, timing or latency data for the survey taker **14** to respond to data choices presented, indication of survey taker distraction or an indication of survey taker misunderstanding of a conjoint environment survey question such as repetition of a same response and the like. According to several aspects the management program **22** receives data from a first camera **48** of the survey taker device **40** or from a second camera **50** of the PC **42**. Camera images and survey taker features may be applied using software of the management program **22** to monitor eye movement of the survey taker **14** to define eye-tracking data indicating survey taker attentiveness or distraction from accomplishing the survey, and visual cues of the survey taker **14** such as facial features representing survey taker distraction, and the like.

(11) Applying input data of the survey taker **14** via the data collection link **46** including camera image data and facial data, the API **44** using the management program **22** monitors the survey taker **14** and identifies when and if a communication signal **52** as a wireless or a wired transfer of data is forwarded to the survey taker device **40** or the PC **42**. A device notification **54** may be generated in response to the communication signal **52**, which may be presented on the interface window **38** or elsewhere on one of the survey taker device **40** or the PC **42** as described below. The device notification **54** may take a form such as a haptic notification **56** or an indicator light **58** such as a blinking or a steady light on the interface window **38** or on one of the survey taker device **40** or the

PC **42** which notifies the survey taker **14** that additional attention needs to be paid to the conjoint environment survey program. If the management program **22** identifies the survey taker **14** has met a predetermined threshold of distraction, boredom, eye-vision wander, response error, inattention or the like, the device notification **54** may also take the form of a rest screen **60**. The rest screen **60** may for example include a non-conjoint environment survey screen, a music video, a relaxation video, or the like which functions for a predetermined period of time to allow the survey taker **14** to regain concentration allowing subsequent completion of the conjoint environment survey. Upon completion of the conjoint environment survey, a survey completion signal **62** is generated which returns the survey taker **14** and forwards results of the conjoint environment survey to the vendor program database home page **12**.

(12) Referring generally to FIGS. **2** through **4** and again to FIG. **1**, according to several aspects, to further retain the interest and concentration of the survey taker **14**, the survey taker interface **36** may be provided with multiple different varieties of choice tasks for selection by the survey taker **14**. With more specific reference to FIG. **2**, a first-choice task screen **64** may define a wedge-choice wheel **66** identifying multiple choice tasks presented in a wedge design of multiple colors having a circular wheel representation. In a first wedge presentation, the wedge-choice wheel **66** may include for example a first wheel segment **68** presenting a first vehicle package of features defining for example a red vehicle color, no sunroof, front-wheel drive, a 180 horsepower (HP) engine, a 7-inch infotainment screen size, no tow package, a 38 mile per gallon (MPG) gas mileage and a 2.4 L engine size.

(13) In a second wedge presentation the wedge-choice wheel **66** may include for example a second wheel segment **70** presenting a second vehicle package of features defining for example an orange vehicle color, no sunroof, front-wheel drive, a 180 HP engine, a 10-inch infotainment screen size, no tow package, a 24 mile per gallon (MPG) gas mileage and a 1.8 L engine size. In a third wedge presentation the wedge-choice wheel **66** may include for example a third wheel segment **72** presenting a third vehicle package of features defining for example a white vehicle color, a sunroof, front-wheel drive, a 180 HP engine, a 10-inch infotainment screen size, a tow package, a 48 mile per gallon (MPG) gas mileage and a 1.6 L engine size. In a fourth wedge presentation the wedge-choice wheel **66** may include for example a fourth wheel segment **74** presenting a fourth vehicle package of features defining for example a grey vehicle color, a sunroof, all-wheel drive, a 200 HP engine, a 7-inch infotainment screen size, no tow package, a 32 mile per gallon (MPG) gas mileage and a 2.4 L engine size.

(14) It is noted the wedge-choice wheel **66** provides one example of feature selections. Multiple other feature selection configurations, quantity of wedges, types and multiple different color combinations may also be used without limiting the scope of the present disclosure.

(15) Upon selection of the exemplary second one of the first, second, third and fourth wheel segments of vehicle packages identified above defining the second wheel segment **70**, a summary screen **76** is presented having the vehicle features identified for the second wheel segment **70** in a column format. A vehicle price **78** is also presented for this vehicle package of features from which the survey taker **14** may further refine a vehicle feature package choice. A question **80** is presented to the survey taker **14** to identify whether the survey taker **14** would or would not purchase the selected vehicle if the vehicle was real and the features of the exemplary selected first wheel segment **68** are all found to be acceptable. In the example provided in FIG. **2**, the survey taker **14** has selected a YES response to the question **80**. The survey taker **14** may then select a back button **82** to return to a previous screen or a next button **84** to move forward with a next vehicle package selection.

(16) Referring to FIG. **3** and again to FIG. **2**, a second-choice task screen **86** may define an angular-choice wheel **88** identifying multiple choice tasks presented having angular rings in a circular wheel design with features presented on multiple wedges and in multiple colors. In a first ring segment **90**, the angular-choice wheel **88** may include for example multiple engine HP

selections arranged in concentric order toward a center of the angular-choice wheel **88**. In a second ring segment **92**, the angular-choice wheel **88** may include for example multiple vehicle average mile-per-gallon selections arranged in concentric order toward a center of the angular-choice wheel **88**.

(17) With continuing reference to FIG. 3, in a third ring segment **96**, the angular-choice wheel **88** may include for example multiple drive type selections arranged in concentric order toward a center of the angular-choice wheel **88**. In a fourth ring segment **98**, the angular-choice wheel **88** may include for example multiple sunroof selections arranged in concentric order toward a center of the angular-choice wheel **88**. In a fifth ring segment **100**, the angular-choice wheel **88** may include for example multiple infotainment screen size selections arranged in concentric order toward a center of the angular-choice wheel **88**. In a sixth ring segment **102**, the angular-choice wheel **88** may include for example multiple towing hitch selections arranged in concentric order toward a center of the angular-choice wheel **88**. In a seventh ring segment **104**, the angular-choice wheel **88** may include for example multiple engine size selections arranged in concentric order toward a center of the angular-choice wheel **88**. Multiple rings such as an exemplary ring **105** provide combined features for the multiple ring segments.

(18) Similar to the summary screen **76** of FIG. 1, a summary screen **106** is presented having the vehicle features identified for the exemplary ring **105** are presented in a column format. A vehicle price **108** is also presented for this vehicle package of features from which the survey taker **14** may further refine a vehicle feature package choice. Similar to FIG. 1, the question **80** is presented to the survey taker **14** and the survey taker **14** may then select the back button **82** to return to a previous screen or the next button **84** to move forward with a next vehicle package selection.

(19) It is noted the angular-choice wheel **88** provides one example of feature selections. Multiple other feature selection configurations, quantity or rings, types and multiple different color combinations may also be used without limiting the scope of the present disclosure.

(20) Referring to FIG. 4 and again to FIGS. 2 through 3, a third-choice task screen **110** defines a choice wheel **112** identifying multiple features in a circular wheel design having different features positioned about a circumference of the choice wheel **112**. Individual rings are provided within the choice wheel **112** distinguishing different feature values for individual ones of the features selected and are successively numbered such as 1, 2, 3, 4 as shown. Individual packages of vehicle feature selections may be identified using multiple different polygons with selected vehicle features identified by contact of individual polygon vertices with individual ones of the annular rings **114**. The multiple feature selections may include for example a fuel economy or MPG **116**, an infotainment package **118**, a range **120**, a price **122** and an engine horsepower (HP) **124**. Feature values selected on individual rings **114** are identified by contact of individual vertices of multiple polygons which provide multiple options based on the survey taker **14** choices.

(21) An exemplary first polygon **126** identifies multiple selected options by contact of individual vertices of the first polygon **126** with individual ones of the rings **114** defining different selected values of the fuel economy or MPG **114**, the infotainment package **118**, the range **120**, the price **122** and the engine horsepower (HP) **124**. An exemplary second polygon **128** identifies multiple selected options by contact of individual vertices of the second polygon **128** with individual ones of the rings **114** defining different selected values of the fuel economy or MPG **114**, the infotainment package **118**, the range **120**, the price **122** and the engine horsepower (HP) **124**. An exemplary third polygon **130** identifies multiple selected options by contact of individual vertices of the third polygon **130** with individual ones of the rings **114** defining different selected values of the fuel economy or MPG **114**, the infotainment package **118**, the range **120**, the price **122** and the engine horsepower (HP) **124**.

(22) Initially, a select vehicle choice field **132** permits the survey taker **14** to make choices of available options between multiple vehicle packages, by selecting a vehicle **1**, a vehicle **2**, a vehicle **3**, a vehicle **4**, a vehicle **5** and the like. For any vehicle selected from the select vehicle choice field

132, one of multiple feature packages **134** listing the features of the selected vehicle are identified by one of the above noted polygons. For example, by selecting vehicle **1** from the vehicle choice field **132** the values identified by the first polygon **126** are identified, by selecting vehicle **2** from the vehicle choice field **132** the values identified by the second polygon **128** are identified, and by selecting vehicle **5** from the vehicle choice field **132** the values identified by the third polygon **130** are identified. Similar to FIG. **1**, the survey taker **14** may then select the back button **82** to return to a previous screen or the next button **84** to move forward with a next vehicle package selection.

(23) It is noted the choice wheel **112** provides one example of feature selections. Multiple other feature selection configurations, quantity or rings, quantity of polygons presented and multiple different color combinations may also be used without limiting the scope of the present disclosure.

(24) According to several aspects, the system and method of the present disclosure is based on a process to manage information flow between vehicle experimental designs, choice task rendering which is device/respondent dependent, choice responses, in-survey engagement, and sensory feedback control mechanisms. A process selects choice task rendering amongst several interface designs optimized for use on a small-scale screen based on a survey taker device, characteristics and identified preferences. The present system design has further interactive options including but not limited to rotation, rollover, zoom in/out and color-highlighting.

(25) Wedge-choice wheels of the present disclosure present choice tasks as wedges in a circular wheel representation with features as angular rings. Angular-choice wheels of the present disclosure present choice tasks as angular rings in a circular wheel representation having vehicle features on wedges. Polygon-choice wheels of the present disclosure present choice tasks as polygons with features on vertices of the polygons.

(26) The present disclosure provides a process or method to measure in-survey interactive behaviors indicative of early survey drop-out propensity including reduced dwell time, number and location of drags, selections and rotations. The present process creates in-survey task breaks in response to measured behaviors to allow time off for a survey taker to relax prior to completing the survey. The location, frequency and content of task breaks may be based on predetermined rules and in-survey measurement levels. The present process also measures survey taker in-survey focused attention and choice task survey taker engagement including a response pattern, for example a survey taker repeatedly selecting a first-choice profile regardless of option content, eye-tracking including prolonged eye movement away from a device interface window, and interactive behaviors indicative of reduced attention including reduced dwell time in responding to choices. The present process induces sensory feedback through visual means including blinking lights, pop-up messages, auditory sounds such as beeps, and haptic vibration mechanisms. The placement and frequency of sensory feedback is determined based on a combination of predetermined rules and strength of in-survey measurements.

(27) The present process and method dynamically determines design elements such as a font size, a number of choice tasks, and sizes of angular, wedge and polygon representations based on physical characteristics of the electronic device used including a screen size; a model of interaction, and use of a mouse or a finger. The present disclosure may be applied to perform conjoint study surveys for vehicle package selection, for eliciting prospective consumer preferences and for forecasting future items such as vehicle packaging.

(28) An interactive conjoint study survey system **10** of the present disclosure offers several advantages. These include a methodology to increase pre-survey response by selecting amongst several interface designs optimized for use on a small-scale screen such as a smart phone-based survey taker device. A control mechanism applied using an application program and an application programming interface increases survey response rate amongst prospective survey takers with a known predispositions toward election to use mobile devices in lieu of portable computers. This provides representativeness of conjoint studies with respect to a target population by capturing a broader selection of prospective survey respondents. The present system and method also improve

in-survey quality by measuring survey taker behavior known to diminish response quality, that impact a completion rate. The present system and method also measures survey taker behaviors indicative of early drop-out, including reduced choice tasks dwell time and tap/drag actions. Early drop-out is mitigated through several sensory feedback control mechanisms, including visual mechanisms including blinking lights and pop-up messages, auditory sounds including beeps, and haptic feedback such as device vibration. A placement and frequency of sensory feedback is determined based on a combination of fixed rules and strength of in-survey measurements.

(29) The present system and method also enhances task comprehension by measuring survey taker behavior indicative of boredom, diminished attention, and reduced choice task engagement. These deleterious behaviors may be mitigated through the control mechanism of a task break, which is an image or exercise unrelated to the conjoint choice exercise that is shown to a survey taker at different points in the conjoint survey. A survey location and a frequency of task breaks are determined using predetermined rules and measure a degree of reduction in focused attention and engagement.

(30) The interactive conjoint study survey system **10** of the present disclosure resolves problems unique to the field of generating and refining vehicle data conjoint surveys including lack of a system and method which may be presented to a survey taker on a reduced screen size such as a smart phone. The claimed interactive conjoint study survey system **10** effectuates an improvement over the known systems and methods by use of a first virtual machine including an application server operating a management program to generate, modify and save a conjoint environment survey content for access by a survey taker. A choice task content list is generated by the management program and is reviewed and completed by the survey taker. A reader program of the first virtual machine identifies multiple survey taker choices selected from the choice task content list to be displayed. A survey taker interface displays the multiple survey taker choices, with the survey taker interface being formatted to conform to an interface window completely displayed on a screen of a survey taker device defining a smart phone. The interactive conjoint study survey system **10** further provides representativeness of conjoint studies with respect to a target population by capturing a broader selection of prospective survey respondents to improve in-survey quality by measuring survey taker behavior known to diminish response quality, that impact a completion rate. The system monitors the survey taker with an application programming interface (API) and identifies when and if a communication signal is forwarded to the survey taker device to help eliminate bad surveys due to survey taker distraction or early dropout. Processing time is reduced and data storage resources are saved by preemptively rejecting bad surveys.

(31) The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. An interactive conjoint study survey system, comprising: a first virtual machine including an application server operating a management program generating, modifying and saving a conjoint environment survey content for access by a survey taker; a choice task content list generated by the management program and reviewed and completed by the survey taker; a reader program of the first virtual machine identifying multiple survey taker choices selected from the choice task content list to be displayed; a survey taker interface displaying the multiple survey taker choices, the survey taker interface being formatted to conform to an interface window completely displayed on a screen of a survey taker device defining a smart phone and the survey taker interface having multiple choice tasks for selection by the survey taker to retain an interest and concentration of the survey taker including displaying the multiple choice tasks on a plurality of circular wheel wedges

presented in a circular wheel design of the survey taker interface; a data collection link including camera image data and facial data wherein applying input data of the survey taker via the data collection link including multiple survey taker camera image data and multiple survey taker facial data, the API using the management program identifies when and if a communication signal as a wireless or a wired transfer of data is forwarded to the survey taker device; a first camera of the survey taker device wherein the management program receives the camera image data and the facial data from the first camera, wherein camera images and survey taker features are applied using the management program to monitor eye movement of the survey taker to define eye-tracking data indicating survey taker attentiveness or distraction from accomplishing the survey, and visual cues of the survey taker including facial features representing survey taker distraction from accomplishing the survey, to identify that the survey taker has met a predetermined threshold of distraction, boredom, eye-vision wander, response error, or inattention; and a device notification generated in response to the communication signal and presented on the interface window or on the survey taker device, the device notification including at least one of a haptic notification and an indicator light including a blinking or a steady light on the interface window or on the survey taker device to notify the survey taker that additional attention needs to be paid to the conjoint environment survey content, and wherein the device notification defines a rest screen when the management program identifies the survey taker has met the predetermined threshold of distraction, boredom, eye-vision wander, response error, or inattention.

2. The interactive conjoint study survey system of claim 1, including a vendor program database collecting information for a conjoint environment survey of data including data provided by a manufacturer enabling the survey taker via a programming function to access the first virtual machine.

3. The interactive vehicle conjoint study survey system of claim 2, including a second virtual machine, the first virtual machine forwarding data requests to and retrieving conjoint environment survey data from the second virtual machine.

4. The interactive conjoint study survey system of claim 3, wherein the second virtual machine includes a database of vehicle developmental data including an attribute list and descriptions of features and options available in a vehicle under development and listed on the choice task content list.

5. The interactive conjoint study survey system of claim 1, including an application programming interface (API) of the first virtual machine receiving information including but not limited to the multiple survey taker choices, a survey taker focus status, survey taker eye-tracking data, timing or latency data for the survey taker to respond to data choices presented, indication of survey taker distraction or an indication of survey taker misunderstanding of a conjoint environment survey question such as repetition of a same response.

6. The interactive conjoint study survey system of claim 1, wherein the survey taker interface is further formatted to conform to being displayed on a viewing screen of a personal computer (PC).

7. A method to perform a vehicle interactive conjoint study survey, comprising: operating a management program to generate, modify and save conjoint environment survey content for access by a survey taker; generating a choice task content list using the management program to be reviewed and completed by the survey taker; monitoring eye movement of the survey taker to define eye-tracking data indicating survey taker attentiveness or distraction from accomplishing the survey using a first camera of the survey taker device, and evaluating visual cues of the survey taker including facial features representing survey taker distraction from accomplishing the survey and the management program to identify that the survey taker has met a predetermined threshold of distraction, boredom, eye-vision wander, response error, or inattention; sending information including multiple survey taker choices, a survey taker focus status, survey taker eye-tracking data, timing or latency data for the survey taker to respond to data choices presented, indication of survey taker distraction or an indication of survey taker misunderstanding of a conjoint

environment survey question including repetition of a same response to an application programming interface (API); displaying the multiple survey taker choices on a survey taker interface of the survey taker device; formatting the survey taker interface to conform to an interface window completely displayed on a screen on the survey taker device defining a smart phone; providing the survey taker interface having multiple choice tasks for selection by the survey taker to retain an interest and concentration of the survey taker including displaying the multiple choice tasks on a plurality of circular wheel wedges presented in a circular wheel design of the survey taker interface; monitoring the survey taker with the API and using the management program to identify when and if a communication signal is forwarded to a survey taker device; generating a device notification in response to the communication signal and presenting the device notification on the survey taker device as a haptic notification or an indicator light including a blinking or a steady light on the survey taker device to notify the survey taker that additional attention needs to be paid to the conjoint environment survey program; and conforming the device notification as a rest screen when the management program identifies the survey taker has met the predetermined threshold of distraction, boredom, eye-vision wander, response error or inattention.

8. The method of claim 7, further including applying camera image data and facial data as input data of the survey taker.

9. The method of claim 7, further including actuating a reader program to identify survey taker choices selected from the choice task content list to be displayed.

10. A method to perform a vehicle interactive conjoint study survey, comprising: operating a management program to generate, modify and save a conjoint environment survey content for access by a survey taker; generating a choice task content list using the management program to be reviewed and completed by the survey taker; monitoring eye movement of the survey taker to define eye-tracking data indicating survey taker attentiveness or distraction from accomplishing a survey using a first camera of the survey taker device, and evaluating visual cues of the survey taker including facial features representing survey taker distraction and using the management program to identify when and if a communication signal is forwarded to a survey taker device and to identify that the survey taker has met a predetermined threshold of distraction, boredom, eye-vision wander, response error, or inattention; sending survey user information including multiple survey taker choices, survey taker selections, a survey taker focus status, survey taker eye-tracking data, timing or latency data for the survey taker to respond to data choices presented, indication of survey taker distraction or an indication of survey taker misunderstanding of a conjoint environment survey question such as repetition of a same response to an application programming interface (API); displaying the multiple survey taker choices on a survey taker interface of the survey taker device; formatting the survey taker interface to conform to an interface window completely displayed on a screen on the survey taker device defining a smart phone; providing the survey taker interface having multiple choice tasks for selection by the survey taker to retain an interest and concentration of the survey taker including displaying the multiple choice tasks on a plurality of circular wheel wedges presented in a circular wheel design of the survey taker interface; monitoring the survey taker with the API and using the management program to identify when and if a communication signal is forwarded to a survey taker device; creating in-survey task breaks in response to measured behaviors of the survey taker to allow time off for the survey taker to relax prior to completing the conjoint environment survey; measuring survey taker in-survey focused attention and choice task survey taker engagement; generating a device notification and presenting the device notification on the survey taker device as a haptic notification or an indicator light including a blinking or a steady light on the survey taker device to notify the survey taker that additional attention needs to be paid to the conjoint environment survey program; and conforming the device notification as a rest screen when the management program identifies the survey taker has met the predetermined threshold of distraction, boredom, eye-vision wander, response error or inattention.

11. The method of claim 5, further including measuring a survey user response pattern including the survey taker repeatedly selecting a first-choice profile regardless of option content, eye-tracking including prolonged eye movement away from a device interface window, and interactive behaviors indicative of reduced attention including reduced dwell time in responding to choices.
